HPE GREENLAKE FOR MICROSOFT AZURE STACK HCI

Reference Architecture

DOCUMENT REVISION HISTORY

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CONTENTS

LIST OF FIGURES

LIST OF TABLES

SOLUTION OVERVIEW

This document provides an architectural overview of HPE GreenLake for Microsoft Azure Stack HCI. This document is intended for the person responsible for architecting and installing the solution.

HPE GreenLake for Microsoft Azure Stack HCI is a cloud-native platform meeting the Microsoft Integrated Systems criteria to provide scalable hyperconverged infrastructure and networking, software, and services in the customer data center. This complete-single rack solution designed for managed services offerings from Hewlett Packard Enterprise is ready to run distributed cloud services from the edge to the cloud providing a consistent operating and management experience on-premises. The platform integrates leading innovations such as cloud-native software combined with leading-edge HPE infrastructure packaged as an end-to-end solution managed and operated by HPE.

The solution is built with standard infrastructure modules designed for Microsoft Azure Stack HCI, meeting the Microsoft Integrated Systems criteria. The goal is to offer a turnkey appliance like experience that is easily integrated in factory on a stack that is continuously validated, offering a full lifecycle management experience, and providing the highest security while delivering cloud experience on-premises.

The solution includes an Integrated Systems Management Plane (ISMP) providing a consolidated management platform for hosting tools necessary for infrastructure and platform management and monitoring. It is a physical server that runs SuSE Linux and provides a KVM-based virtualized environment for running on-premises management applications and gateways for cloud-based management applications.

HPE GreenLake for Microsoft Azure Stack HCI has the following features:

- Hosts virtualized and containerized workloads and their storage in a hybrid environment that is delivered through the combination of on-premises infrastructure and Microsoft Azure cloud services.
- Delivered as a Microsoft Azure service and billed on a per-core basis.
- Creates a hyperconverged cluster that features Hyper-V, Storage Spaces Direct for hyperconverged storage and Software-Defined Networking. For more information, refer to https://docs.microsoft.com/en-in/azure-stack/hci/overview.

Microsoft Azure Stack HCI, version 21H2 is the latest offering intended for on-premises clusters running virtualized workloads, with hybrid-cloud connections built-in. At its core, Microsoft Azure Stack HCI is a solution that combines the following:

- Azure Stack HCI operating system
- Validated hardware
- Azure hybrid services
- Windows Admin Center
- Hyper-V-based compute resources
- Storage Spaces Direct-based virtualized storage
- SDN-based virtualized networking using Network Controller (SDN is not part of the default solution that HPE GreenLake offers)

The HPE GreenLake for Microsoft Azure Stack HCI solution offers:

- Minimum of two and maximum of 16 nodes of HPE ProLiant DL380 Gen10 Plus server is made available in six instances. Four instances of all NVMe storage, and 1 instance of All Flash SSDs, with and without GPU. Each of these instances come pre-defined with raw capacity per node, type of CPUs, and amount of memory.
- Management for all component with the HPE ProLiant DL325 Gen10 Plus v2 server along with metering and billing services to provide a complete as-a-service experience.
- Networking using Aruba 8325 and 6300M switches
- Single vendor experience for the entire stack from factory build to onsite implementation and support.
- Deliver as-a-service offering to meet customer needs:
  - Managed Infrastructure with HPE managing the hardware components.

- Managed Platform with HPE managing hardware and Hypervisor OS.
- Fully managed service with HPE managing the hardware, Hypervisor OS, and application hardware.

The rack with minimum and maximum nodes that the customer can order is depicted in Figure 1.

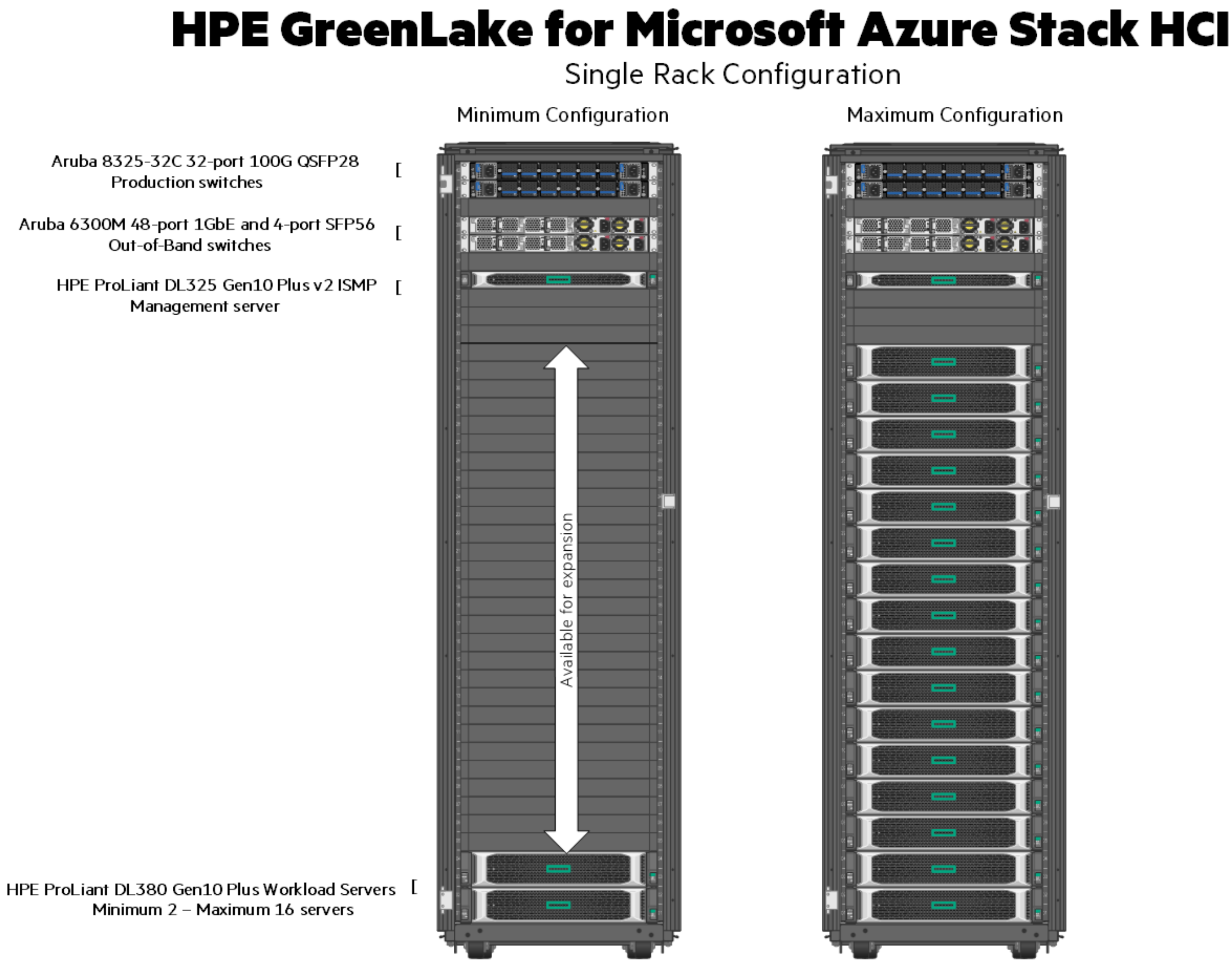

FIGURE 1. HPE GreenLake for Microsoft Azure Stack HCI architecture

SOLUTION COMPONENTS

Microsoft Integrated Systems validation and partner requirements

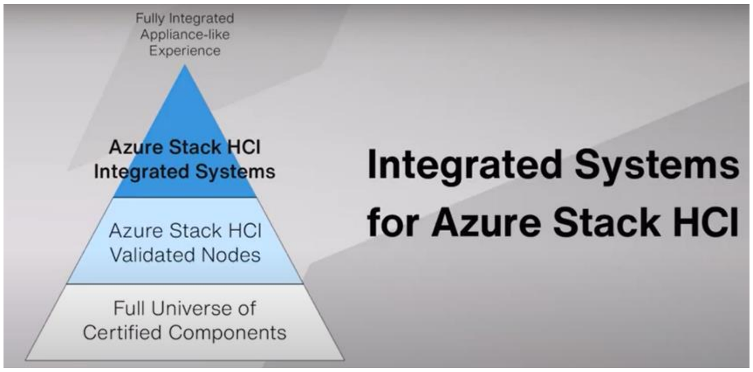

FIGURE 2. Microsoft Azure Stack HCI integrated systems

Microsoft has several requirements in place for Hewlett Packard Enterprise to certify validated node configurations:

- The system (server) must have passed the Windows Server 2022 certification and have the Software Defined Data Center (SDDC) Premium Additional Qualifications (AQ). The system must be listed in the Windows Server Catalog with the SDDC Premium AQ.
- The storage HBA must have passed the Windows Server 2022 certification and support pass-through to the disks (not RAID). The HBA must be listed in the Windows Server Catalog.
- The Network Interface Card must have passed the Windows Server 2022 certification and have the SDDC AQ. This can be either the SDDC Standard or SDDC Premium AQ. The NIC must be listed in the Windows Server Catalog with the SDDC AQ.
- The storage devices must have passed the Windows Server 2022 certification and be listed in the Windows Server Catalog.
- The solution must pass the required validation tests in the HLK targeting an Azure Stack HCI cluster (testing details are called out later)
- S2D: Basic Verification Tests
- S2D: Stress Tests
- Private Cloud Simulator – Storage Spaces Direct (Min): S2D tests need to be run on the servers that are to be certified by Microsoft. After submission, Microsoft evaluates the results before approving the configuration.
- Affirm that customers may receive up to five years of hardware warranty support from the time of customer's purchase for listed Validated Nodes and Integrated Systems while promoted in the catalog.

The HPE server validated node configurations are used to provide an integrated system experience. This is made possible by installing Microsoft Azure Stack HCI operating system in the factory, offering the servers to be ordered as a multi-node solution, and a joint support agreement between Microsoft and Hewlett Packard Enterprise for a seamless support experience for customers.

In addition to requirements that are in place for validated nodes, Hewlett Packard Enterprise met the requirements listed to qualify as a Microsoft Azure Stack HCI integrated systems partner.

- TPM2.0 installed and enabled (the solution must ship with a TPM 2.0 certified module)
- Secure boot enabled
- VBS enabled
- HVCI enabled
- Kernel DMA pre-boot protection enabled

- System Guard Secure Launch enabled
  - The system must have the Secured-core Server AQ and be listed in the Windows Server Catalog if it is a 3rd Gen Intel Xeon SP or 3rd Gen AMD EPYC, or newer processor (does not apply to drop-in upgrade from 2nd Gen AMD EPYC motherboard).
  - Kernel Soft Reboot – Optional for HCI 21H2 (Not mandatory for HCI 21H2, but encouraged by Microsoft)
  - The solution must pass these additional validation tests using the Hardware Lab Kit (HLK)
    - Kernel Soft Reboot Device
    - Kernel Soft Reboot Functional
    - Kernel Soft Reboot Reliability
- Creation and support of a Windows Admin Center plug-in for deployment and update
- Pre-install of the Microsoft Azure Stack HCI operating system
- Highest component quality in the solution, tracked through EPSO quality review process
- Solution must be ordered as a multi-node solution
- Joint support agreement between the solution builder and Microsoft
- The solution builder will host a 4-node solution for ongoing validation of firmware and driver updates prior to public release
- The system must have a unique SMBIOS name that is not used by a Validated Node.

Validated nodes

HPE GreenLake for Microsoft Azure Stack HCI offers a choice of ready-to-go validated solution configurations, based on specific HPE ProLiant Servers and components that were tested, optimized, and validated with the Microsoft Azure Stack HCI operating system to deliver reliable solid performance and high availability. These validated node configurations are periodically updated at https://www.hpe.com/us/en/alliance/microsoft/azurestackhci.html.

Optional features

Validated nodes can optionally support additional features.

- Secured-core Server
  - The system must have the Secured-core Server AQ and be listed in the Windows Server Catalog if it is a 3rd Gen Intel® Xeon® SP or 3rd Gen AMD EPYC, or newer processor (This does not apply to drop-in upgrade from 2nd Gen AMD EPYC motherboard).
  - The Technical use case document (aka “Badge” document) for Secured-core Server must also be completed and posted before the solution can be listed in the Azure Stack HCI Catalog (on supporting platform).
- Kernel Soft Reboot
  - The solution must pass these additional validation tests using the Hardware Lab Kit (HLK):

    Kernel Soft Reboot Device

    Kernel Soft Reboot Functional

    Kernel Soft Reboot Reliability

Workload modules

Compute building blocks for the solution are built using HPE ProLiant DL380 Gen10 Plus servers.

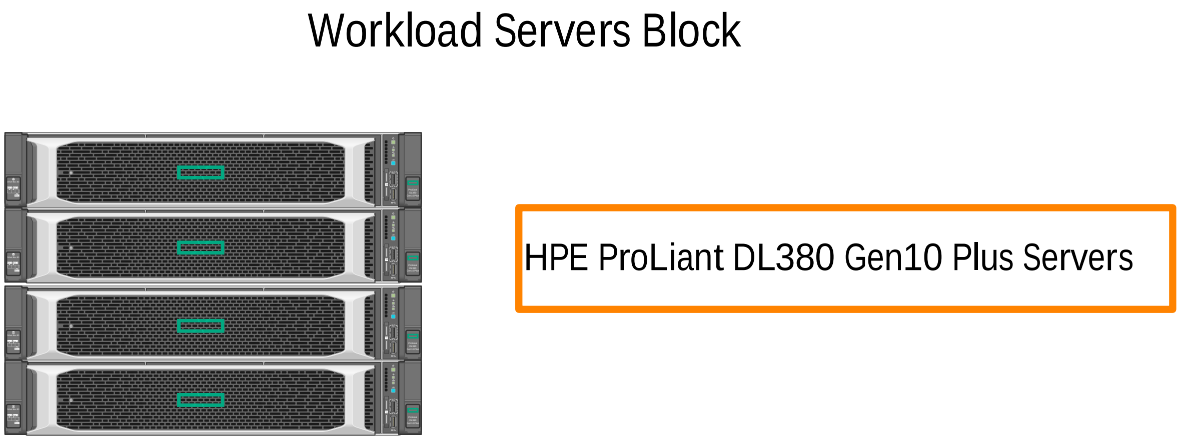

FIGURE 3. Solution workload servers

Depending on the type of workloads, HPE GreenLake offers SSD, SSD with GPU, NVMe, and Hybrid configurations along with 10/25 GbE network interfaces. The Microsoft Azure Stack HCI operating system is factory installed along with additional configurations in the BIOS and on the operating system.

Storage module

Microsoft Azure Stack HCI operating system uses Storage Spaces Direct (S2D), a hyperconverged infrastructure that uses servers with locally attached drives to create highly available and scalable software-defined storage. There are different types of deployment possibilities using a variety of drive types.

- HPE GreenLake offers All-flash SSD and All-NVMe drive types.
- The Microsoft Azure Stack HCI operating system is set up on a RAID 1 attached to Broadcom MegaRAID MR216i-a x16 12 G Controller for HPE Gen10 Plus.
- Drives used for S2D volumes are attached to a Broadcom MegaRAID MR216i-p x16 lanes No Cache NVMe/SAS 12 G Controller for HPE Gen10 Plus.
- Drives used for S2D volumes are direct attach without a controller in case of NVMe drives and are set up by default to work in HBA mode, which is a requirement for HPE GreenLake for Microsoft Azure Stack HCI.

NVMe Configurations

There are four different NVMe configurations offered, as shown in Table 1. The instance equivalence to these offerings is shown in Table 2.

TABLE 1. NVMe offerings

Raw Capacity	25.6 TB	51.2TB	102.4 TB	102.4 TB
Memory	384 GB	768 GB	768 GB	1536 GB
CPU	2x 24 Core Gold	2x 24 Core Gold	2x 24 Core Gold	2x 24 Core Gold
Network	2x 10/25 Gbps	2x 10/25 Gbps	3x 10/25 Gbps	3x 10/25 Gbps

TABLE 2. NVMe instances offered for HPE GreenLake for Microsoft Azure Stack HCI

Instance type family	Instance Code	Storage type	Storage	Physical Cores	Memory	GPU
Compute Storage Optimized	S2i.N25	NVMe	25.6 TB	48	384 GB	0
Compute Storage Optimized	S2i.N51d	NVMe	51.2TB	48	768 GB	0
Compute Storage Optimized	S2i.N102d	NVMe	102.4 TB	48	768 GB	0
Compute Storage Optimized	S2i.N102q	NVMe	102.4 TB	48	1536 GB	0

SSD Configurations

Customers can choose from two different SSD configurations one with NVIDIA A40 GPU's as shown in Table 3.

TABLE 3. SSD configurations

Raw Capacity	153.6 TB	61.44 TB
Memory	768 GB	1024 GB
CPU	2x 24 Core Intel Xeon Gold	2x 24 Core Intel Xeon Gold
Network	2x 10/25 Gbps	2x 10/25 Gbps
GPUs	0	2x A40 per server

TABLE 4. SSD instances offered for HPE GreenLake for Microsoft Azure Stack HCI

Instance type family	Instance Code	Storage type	Storage	Physical Cores	Memory	GPU
Compute Storage Optimized	S2i.Z153d	SSD	153.6 TB	48	768 GB	0
Compute Storage Optimized	s2iv.Z61d	SSD	61.44 TB	48	1024 GB	2 NVIDIA A40

TABLE 4. Drives used for caching, capacity and caching behavior details

Deployment	Capacity drives	Cache drives	Cache behavior
All SSD	SSD	None Optional: Configure manually	Write only when configured
All NVMe	NVMe	None Optional: Configure manually	Write only when configured

Network

Networkmodules

Networking for the solution is from a pair of Aruba 6300M and 8325 switches.

- Management connectivity in this solution is via a pair of Aruba 6300M 48-port 1GbE and 4-port SFP56 switches
- Leaf connectivity for ToR switches connecting to the data center switches is served via a pair of Aruba 8325-32C 32-port 100G QSFP+/QSFP28 switches

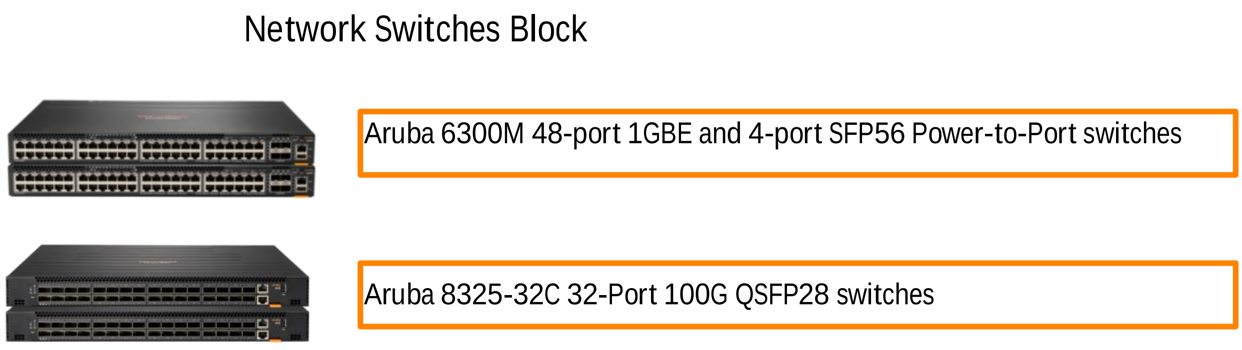

FIGURE 4. Network switches

Network design

Figure 5 represents the high-level design of the connectivity across the solution.

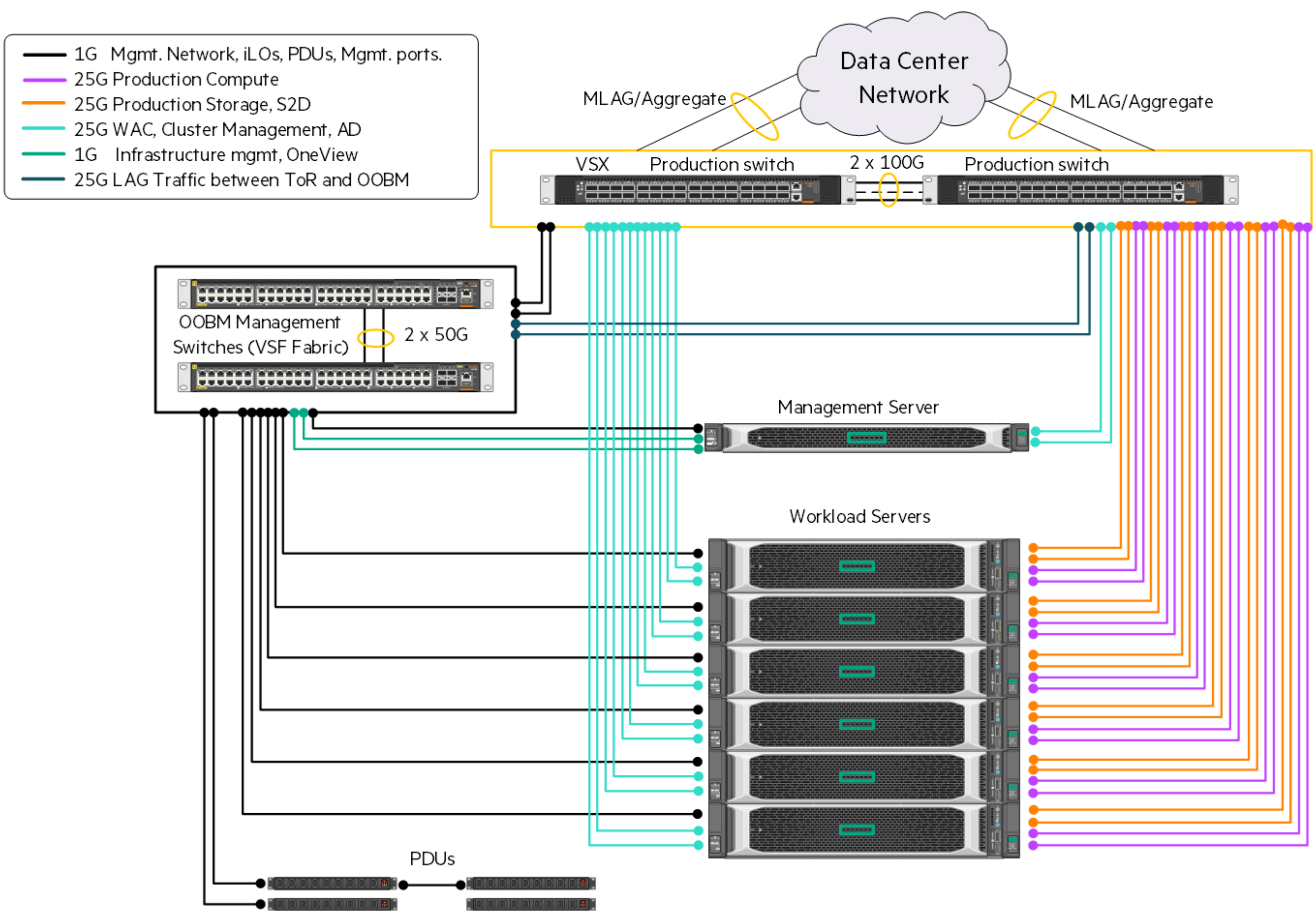

FIGURE 5. High-level design of connectivity across the solution

Detailed individual level design can be found in the *HPE GreenLake for Microsoft Azure Stack HCI Network Guide*. Servers that are configured with high capacity NVMe drives are powered with an additional network card to provide more bandwidth.

Software

System Software

SUSE Linux Enterprise Server is the Operating System. KVM is used as the virtualization solution for creating virtual appliances and VMs required for services application. VM Guests (virtual machines), virtual storage and networks can be managed with libvirt-based tools. The libvirt library provides an API to manage VM Guests via a graphical user interface as well as a command line interface.

Software and firmware matrix

The solution is a validated configuration, and by Microsoft's definition of integrated systems, Hewlett Packard Enterprise must provide ongoing validation of firmware and driver updates prior to public release. Hewlett Packard Enterprise must also test the firmware and drivers on the annual update and agree to support the hardware platform for 5+ years for the solution to stay active on the catalog.

Firmware and driver updates of the solution modules should be updated via Support Pack for ProLiant (SPP). Offline updates via SPP ISO image supports firmware updates, and online installation of the SPP allows the update of both system firmware and OS drivers. HPE Cluster Deployment Snap-in supports firmware and driver updates at the time of cluster creation during the factory integration process. Cluster Update Snap-in will enable seamless updates for ongoing management and operations via a Windows feature called Cluster-Aware updating (CAU) for updating firmware and drivers on servers that are already part of a cluster.

Software includes appliances such as HPE OneView Appliance for monitoring HPE ProLiant DL380 Gen10 Plus servers, WAC and WAC extensions, as well as various HPE support tools.

Management

Hewlett Packard Enterprise offers three levels of management services for HPE GreenLake for Microsoft Azure Stack HCI:

- Managed infrastructure to handle day-to-day operations and management of hardware
- Managed platform to handle the virtualization platform layer
- Managed workload which includes responsibility to run workloads

HPE GreenLake Management Services (GMS) provides day-to-day operational support services that includes 24x7 monitoring and operation of the management hardware, virtualization layer, and applications based on the managed service level being offered.

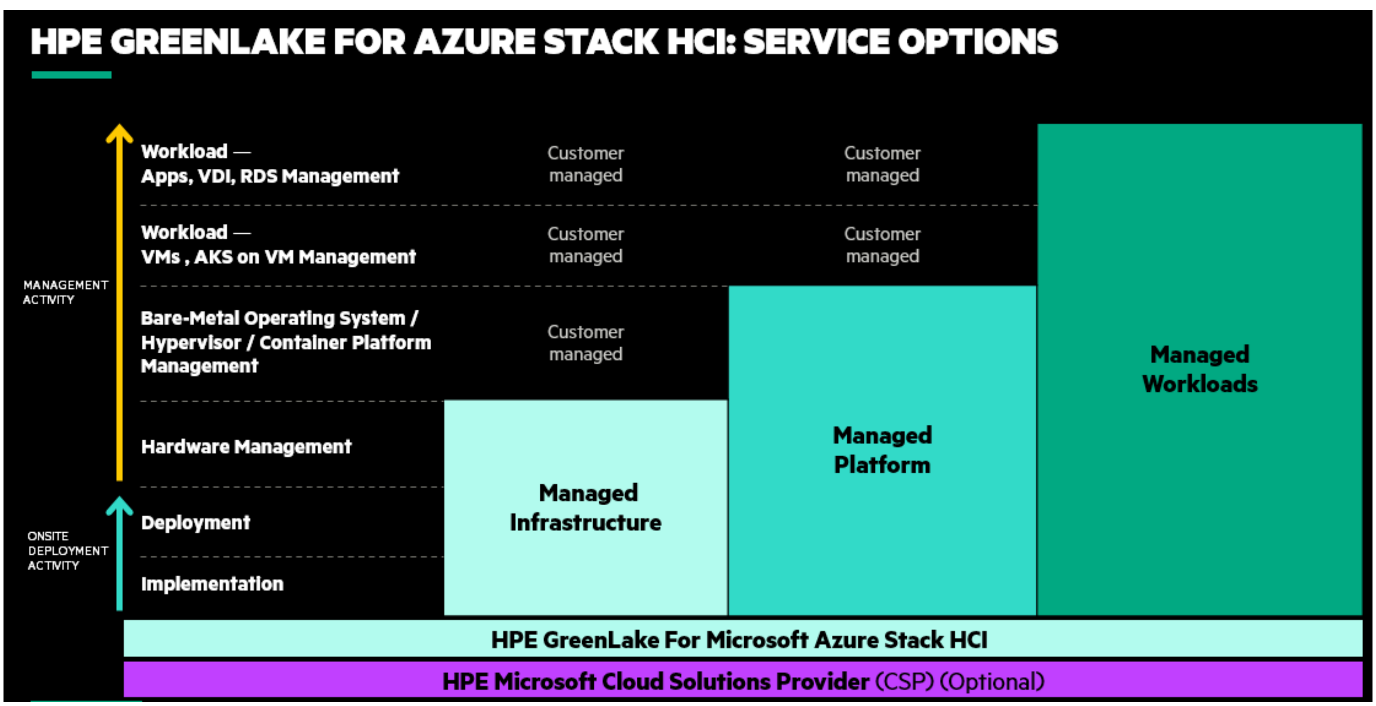

FIGURE 6. Customer managed options

The Management Server

The solution includes a consolidated management platform, Integrated Systems Management Plane (ISMP), for on-premises monitoring and management of the solution infrastructure, and Microsoft Azure Stack HCI. The ISMP was designed to run a set of management applications hosted over a small footprint of physical hardware with required resiliency, availability, and robustness. Metrics and reporting for the infrastructure are through applications running on the management plane.

Management Server

HPE ProLiant DL325 Gen10 Plus v2 Server

FIGURE 7. ISMP Server

Features of the management server:

- HPE ProLiant DL325 Gen10 Plus v2 server with:
  - AMD EPYC 7543P 2.8GHz 32-core 225W Processor
  - Internal storage disks provide storage for boot volume and VMs.
  - Two Ethernet adapters are included to support the networking requirements for the solution.
- A consolidated management platform for hosting tools necessary for infrastructure and platform management, and monitoring.
- Runs SUSE Linux and provides a KVM-based virtualized environment for running on-premises management applications and gateways for cloud-based management applications.
- Hosts server management tools, switch management tools, Microsoft Windows Admin Center, HPE GreenLake metering scripts, HPE GreenLake remote monitoring and management tools.
- Used for initial deployment bootstrap as well as ongoing hardware monitoring, management and support of the solution rack.
- Does not contain customer workload data.

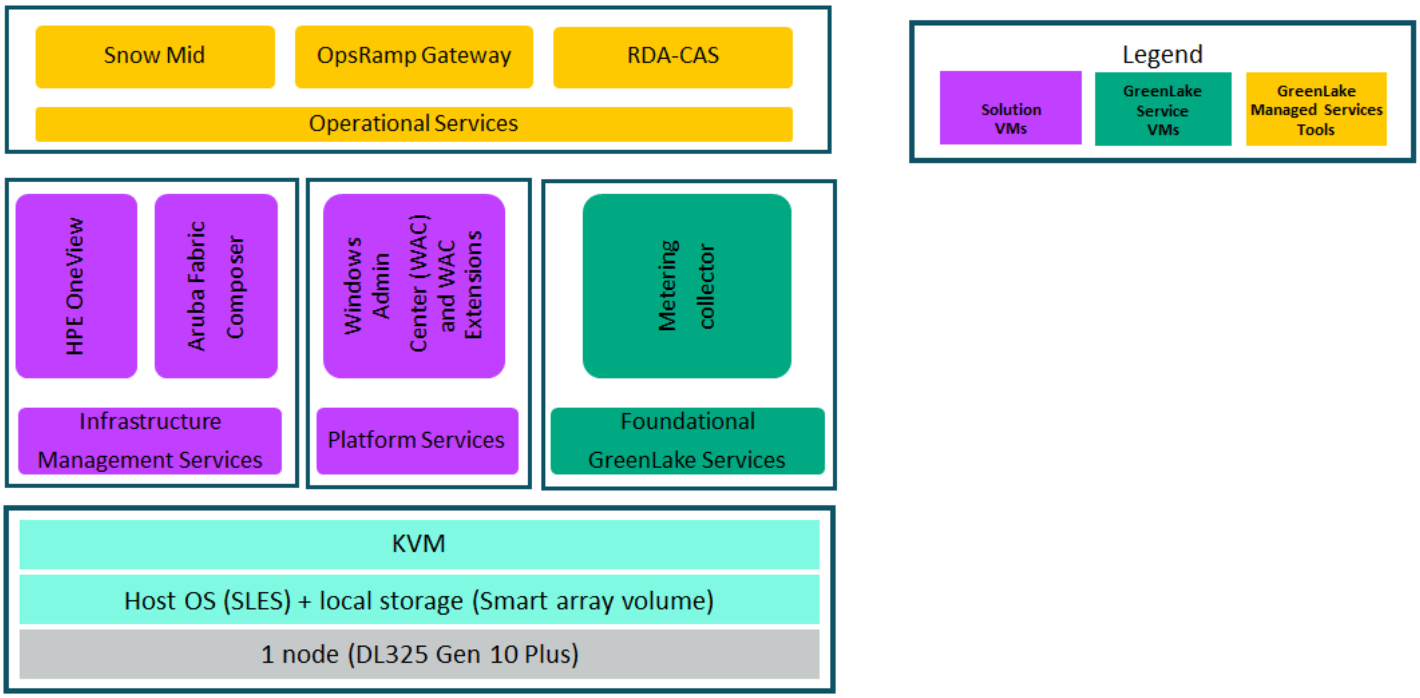

FIGURE 8. Management plane architecture

Table 6 further summarizes functions of each of the applications that are hosted on the management server.

TABLE 5. Functions of applications hosted on the management server

Application/Service	Function
HPE OneView	Monitors all the ASHCI nodes and IS nodes
Aruba Fabric Composer	Handles day-to-day operations of network switches
Windows Admin Center	Manage and monitor ASHCI servers
GL Metering Scripts	Usage Metering and reporting
RDA-CAS	Secured remote access for ITOC admin to access the system
ServiceNow MidServer	Service Now service component used by GMS to automate incident management for the solution
OpsRamp agent	Gateway service used by GMS to monitor and manage physical and virtual resources

The operational view is outlined in Figure 9.

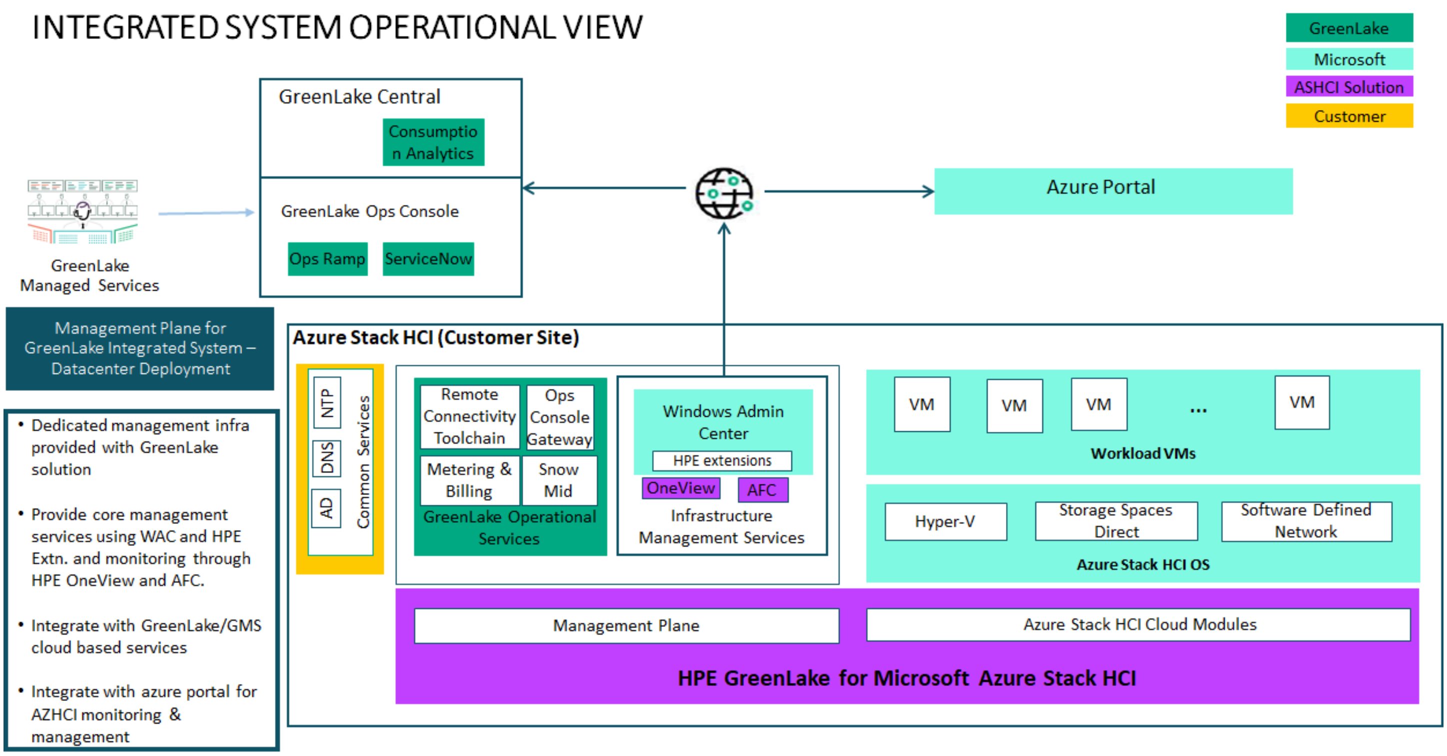

FIGURE 9. Solution Ooerational view

Security features

Microsoft requires the following secured-core features to be enabled by default. These are enabled at the factory and do not require additional change in configurations.

- TPM2.0 installed and enabled
- Secure boot enabled
- VBS enabled
- HVCI enabled
- Kernel DMA pre-boot protection enabled
- System Guard Secure Launch enabled

TABLE 6. Single rack infrastructure minimum and maximum limits

Solution Infrastructure	Minimum	Maximum	Model
Rack	1	1	HPE 42U 600mmx1200mm G2 Enterprise Shock Rack
Top of Rack Switches	2	2	Aruba 8325
Out of band Management switches	2	2	Aruba 6300M
ASHCI Integrated Systems	2	16	HPE ProLiant DL380 Gen10 Plus
ASHCI Integrated System Management Plane	1	1	HPE ProLiant DL325 Gen10 Plus v2

WORKFLOWS

Solution order and delivery workflow

Figure 10 shows the order fulfillment process.

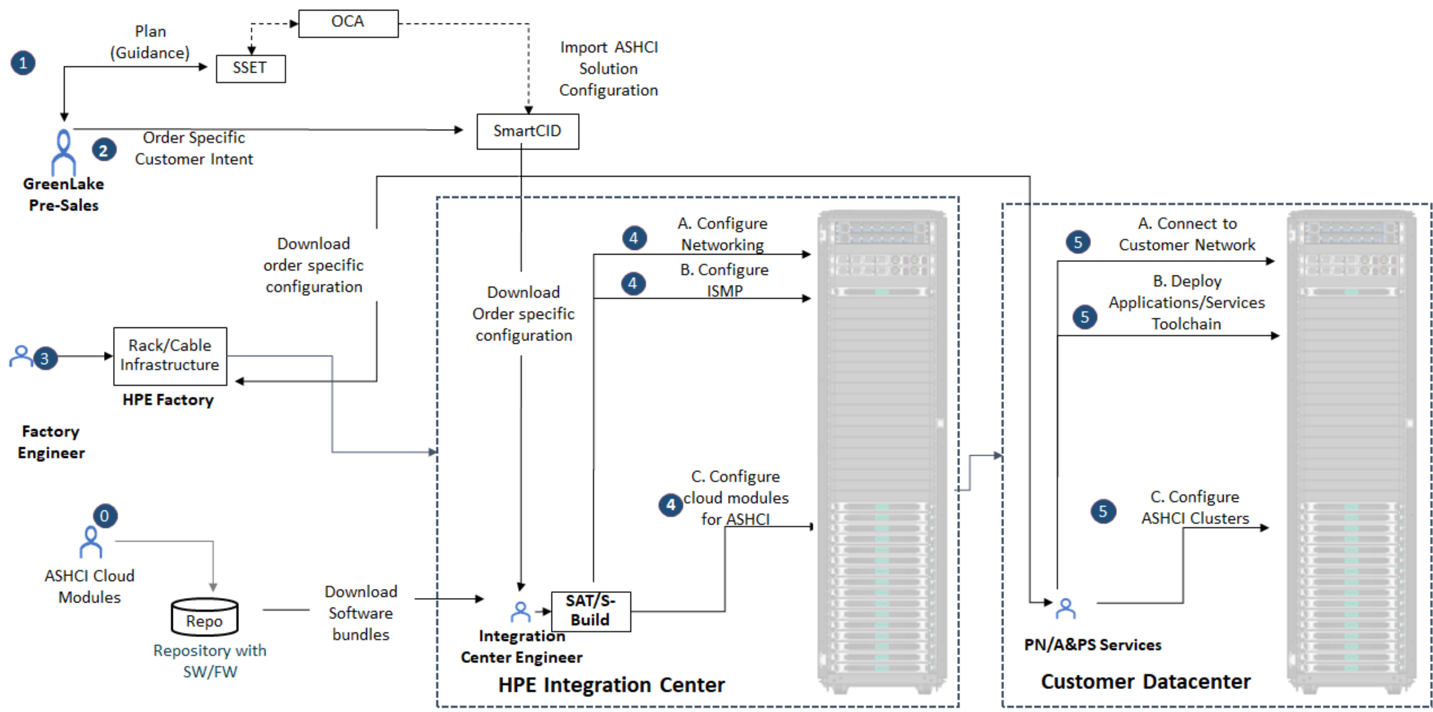

FIGURE 10. Order fulfillment process

Order

Planning guidance for the solution is available through the Solution Sales Enablement Tool (SSET). SSET provides integration with the OCA ordering system. The order specific BOM is auto generated based on the configuration planned in SSET, providing a seamless ordering experience. SSET can be accessed using this link: https://sset.ext.hpe.com.

Build (Factory)

The solution will be pre-configured with customer environment specific details such as VLAN IDs, IP addresses in Factory Integration Center. Details for customer intent are collected through Smart CID and used by the Factory Integration Center engineers for solution system configuration. SCID can be accessed using this link: https://c4w24992.itcs.hpe.com/xFrameworkCIDRestAPI/index-new.html.

Integrate (Onsite)

HPE PointNext Services integrates the solution into the customer datacenter.

Factory workflow

Figure 11 shows tasks done in the factory to build the solution. Detailed instructions are documented in the *HPE GreenLake for Microsoft Azure Stack HCI Factory Integration Guide*.

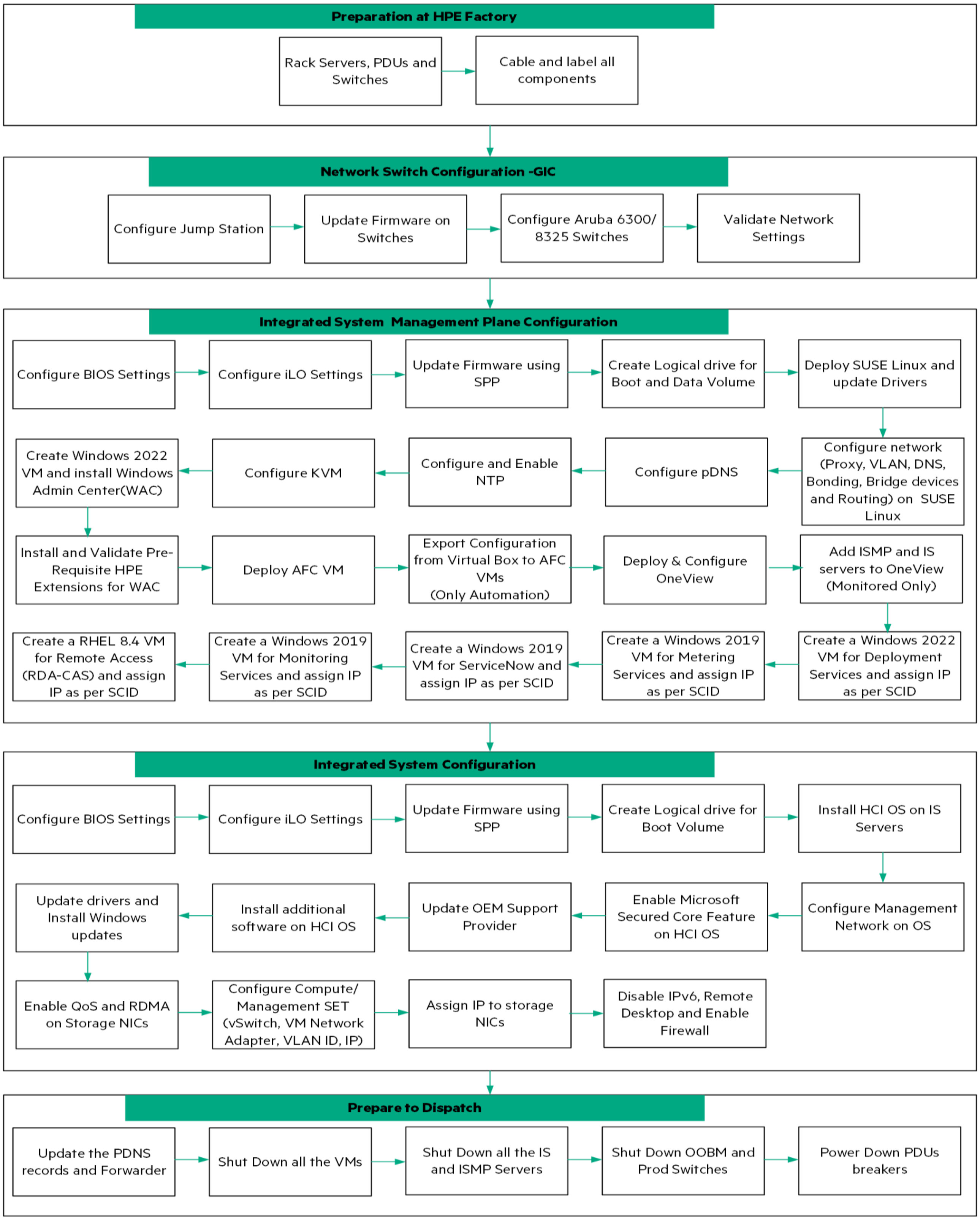

FIGURE 11. Factory workflow

Onsite deployment workflow

The workflow shown in Figure 12 is a representation of different tasks done onsite at customer premises to build the systems. Detailed instructions are documented in the *HPE GreenLake for Microsoft Azure Stack HCI Onsite Deployment Guide*.

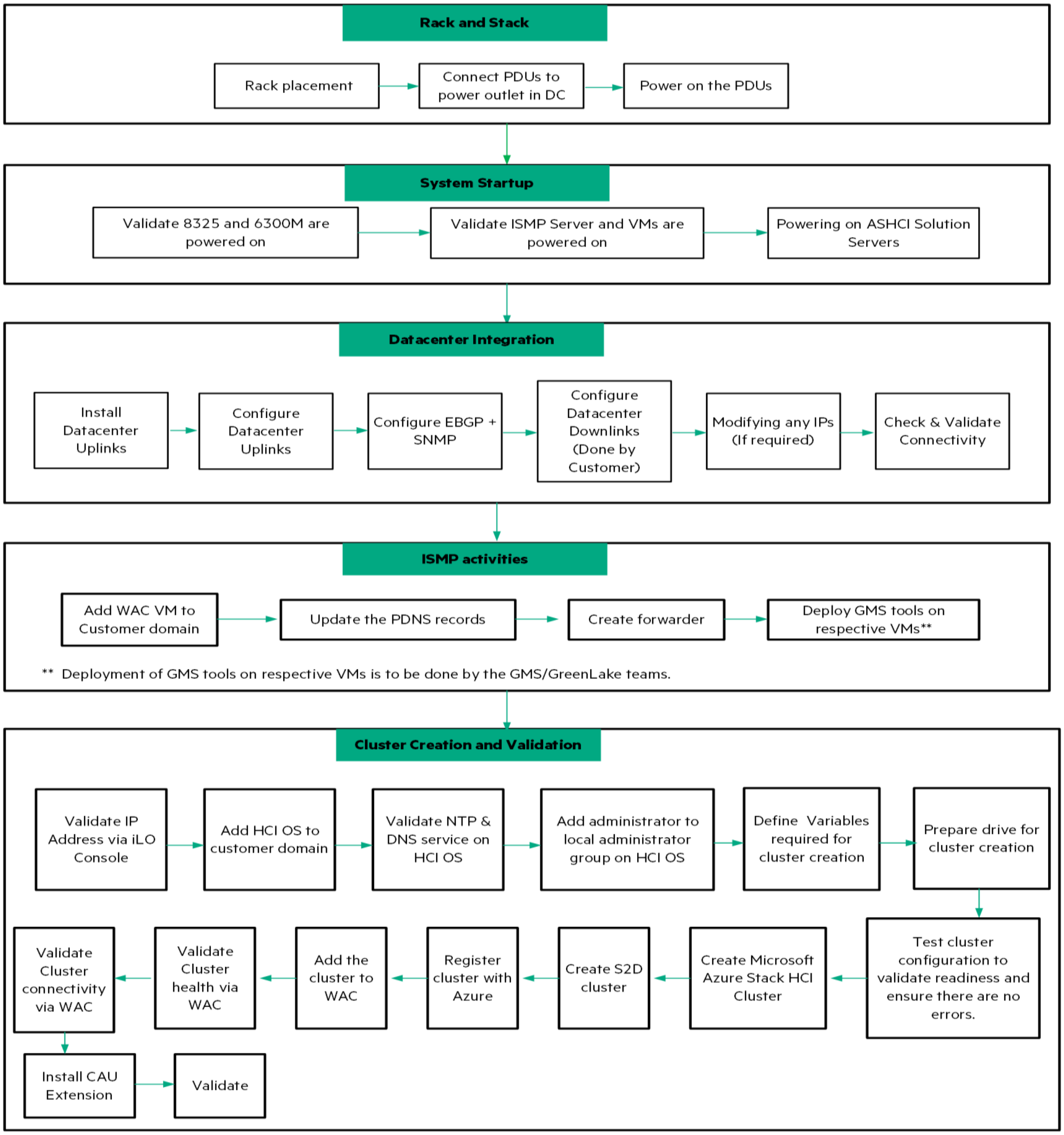

FIGURE 12. Onsite deployment workflow

SOLUTION DOCUMENTATION

The following lists the documents to be used for deploying HPE GreenLake for Microsoft Azure Stack HCI.

TABLE 7. Solution documentation

Document title	Document purpose
HPE GreenLake for Microsoft Azure Stack HCI Reference Architecture	Provides an overview of all the elements of the HPE GreenLake for Microsoft Azure Stack HCI solution.
HPE GreenLake for Microsoft Azure Stack HCI Rack Diagrams	Sample rack elevations
HPE GreenLake for Microsoft Azure Stack HCI Power Guide	Covers supported PDUs and PDU cabling layout
HPE GreenLake for Microsoft Azure Stack HCI Cable Labeling Reference	Cable labeling information
HPE GreenLake for Microsoft Azure Stack HCI Cabling Guide	Sample point-to-point cabling
HPE GreenLake for Microsoft Azure Stack HCI Ethernet Diagram	Provides network design information
HPE GreenLake for Microsoft Azure Stack HCI Server Build Guide	Provides server build details such as memory configurations, drives, drive cages, and slotting information for PCIe cards
HPE GreenLake for Microsoft Azure Stack HCI Checkout Procedure	Checklist for factory integration and onsite deployment.
HPE GreenLake for Microsoft Azure Stack HCI Network Guide	Provides network design information
HPE GreenLake for Microsoft Azure Stack HCI Factory Integration Guide	Provides instructions for factory integration
HPE GreenLake for Microsoft Azure Stack HCI Onsite Deployment Guide	Provides procedures for onsite solution integration
HPE GreenLake for Microsoft Azure Stack HCI Server Security Guide	Details solution security features, along with Microsoft secure core features
HPE GreenLake for Microsoft Azure Stack HCI Firmware and Software Compatibility Matrix	The matrix lists all the firmware and software versions along with the operating system version required to keep the solution up-to-date
HPE GreenLake for Microsoft Azure Stack HCI Management Guide	Provides instructions for operations and management tasks
HPE GreenLake for Microsoft Azure Stack HCI Configuration Worksheet	Worksheet to record customer information required to configure the solution
HPE GreenLake for Microsoft Azure Stack HCI Startup and Shutdown Guide	Start up and shutdown sequence for factory and onsite services personnel

RESOURCES AND ADDITIONAL LINKS

HPE GreenLake for Microsoft Azure Stack HCI documentation
hpe.com/support/HPE-GreenLake-MS-AzureStack-HCI
Log in to HPE Support Center as an HPE Employee to view the internal documents.

HPE Reference Architectures
https://www.hpe.com/docs/reference-architecture

HPE Servers
hpe.com/servers

HPE Storage
hpe.com/storage

HPE Networking
hpe.com/networking

HPE GreenLake Advisory and Professional Services
https://www.hpe.com/us/en/services/consulting.html

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